

Summary

- Industry** End-to-end experience in designing and deploying xx-billion-parameter models in production. Expertise in bridging theoretical analysis with practical problem-solving in industrial settings. 3 years at Google Core ML & 9 months at MIT-IBM Watson AI Lab
- Research** 21 publications (9 as first-author, 7 as corresponding-author), 2 patents, and 390+ citations
- Expertise** Machine Learning (large-small model collaboration, personalization, recommendation systems, LLM, learning to rank), Trustworthy AI (differential privacy, robustness, explainability), Computational Social Choice, and Theoretical Computer Science.

Education

- 1/2018 – 5/2023 **Ph.D. in Computer Science**, Rensselaer Polytechnic Institute (RPI) Troy, NY, USA
Thesis: *Group Decision Makings from Partial Preferences*
- 8/2015 – 5/2018 **M.Eng. in Material Engineering**, Rensselaer Polytechnic Institute Troy, NY, USA
Received *Presidential Graduate Research Fellowship*
- 8/2010 – 5/2014 **B.S. in Mathematics and Physics**, Tsinghua University Beijing, China
Minor in Computer Technology, and in *Academic Talent Program*

Experience (all full-time)

- 7/2023 – 1/2026 **Machine Learning Engineer** at **Google Core ML** Mountain View, CA, USA
Project 1: SplitFormer, A Low-Latency, Splittable Transformer Architecture
- Conducted a systematic theoretical analysis demonstrating that prevalent approaches (e.g., FlashAttention) cannot provide meaningful latency reduction for the target model; proposed SplitFormer and proved its theoretical lower bound of 90% inference latency reduction.
 - Held end-to-end responsibility for architecture design, implementation, training, and production deployment of SplitFormer, including integration into the *Google Shopping* recommendation system.
 - Achieved 3× model parameters and 5% relative accuracy improvement while reducing compute consumption by 30%, corresponding to multi-million-dollar estimated annual revenue contribution on *Google Shopping*.
- Project 2: User Behavior Summarization via Tokenization
- Designed and deployed an end-to-end tokenization pipeline that maps user behavior summaries to LLM-compatible token representations, covering model training, token-storage, inference, and low-latency serving.
 - Reduced storage footprint by approximately 60% and training cost by 25%, compared to raw embedding baselines.
- Project 3: Large Encoder Model for User Behavior Prediction & Summarization
- Designed and productionized its Time Embedding module – from design to system integration – improving prediction accuracy by 17% and summarization accuracy by 4% (relative).
 - Contributed to the full model lifecycle, including architecture design, data processing and feature engineering, pre-training and post-training, and online serving.
- 5/2022 – 8/2022 **Research Intern** at **Google Core ML** Mountain View, CA, USA
Project: Unbiased Learning to Rank with a More Accurate Position Bias Estimator
- Proposed a novel probabilistic model for estimating and correcting position bias in recommendation systems; designed and trained the corresponding system.
 - Outperformed the previous state-of-the-art by approximately 6% in prediction accuracy.

- 5/2019 – 8/2019 **Visiting Scholar at MIT-IBM Watson AI Lab** Cambridge, MA, USA
 and Project: Certifiably Robust Interpretation via Rényi Differential Privacy
- 9/2018 – 1/2019
- Proposed the first algorithm with provable robustness against ℓ_∞ -norm adversarial attacks.
 - Achieved 12% improvement in robustness over the state-of-the-art, with no additional computational cost and higher accuracy.
 - Results published as a top-tier paper and two U.S. patents.

Skills

- Programming Python, C/C++, TensorFlow, PyTorch, MATLAB
- Design Large-small model collaboration, personalization, LLM, complexity analysis, user modeling, MCMC, privacy analysis, robustness analysis, model interpretability, smoothed analysis

Selected Research Outputs (* denotes corresponding author)

- AIJ & AAAI **Certifiably Robust Interpretation via Rényi Differential Privacy**
Ao Liu*, Xiaoyu Chen, Sijia Liu, Lirong Xia, and Chuang Gan
- TMLR **Smoothed Differential Privacy**
Ao Liu*, Yu-Xiang Wang, and Lirong Xia
- AAAI (oral) **Near-Neighbor Methods in Random Preference Completion**
Ao Liu*, Qiong Wu, Zhenming Liu, and Lirong Xia
- AAAI (oral) **Learning Plackett-Luce Mixture from Partial Preferences**
Ao Liu*, Zhibing Zhao, Chao Liao, Pinyan Lu, and Lirong Xia
- AAAI (spotlight) **The Semi-Random Likelihood of Doctrinal Paradoxes**
Ao Liu* and Lirong Xia
- UAI **Accelerating Voting by Quantum Computation**
Ao Liu*, Qishen Han, Lirong Xia, and Nengkun Yu
- ETRA (oral) **Differential Privacy for Eye-Tracking Data**
Ao Liu*, Lirong Xia, Andrew Duchowski, Reynold Bailey, Kenneth Holmqvist, and Eakta Jain
- UAI (oral) **How Private Are Commonly-Used Voting Rules?**
Ao Liu, Yun Lu*, Lirong Xia, and Vassilis Zikas
- Polymer Physics **Simulation of Pulse Responses of Lithium Salt-Doped Poly-Ethyleneoxide**
Ao Liu, F. Zeng*, Y. Hu, S. Lu, W. Dong, X. Li, C. Chang, and D. Guo
- US Patent **Certifiably Robust Interpretation**
Ao Liu, Sijia Liu, Bo Wu, Lirong Xia, Qi Cheng Li, and Chuang Gan
- US Patent **Interpretation Maps with Guaranteed Robustness**
Ao Liu, Sijia Liu, Abhishek Bhandwalder, Chuang Gan, Lirong Xia, and Qi Cheng Li
- See all 21 publications and 2 patents at [[Personal Website](#)] [[Google Scholar](#)]

Academic Impact

- Reviewer Journal: Information Sciences, ACM ToIS, TMLR, DMLR, IPA, and Sankhya B
 Conference: NeurIPS, ICML, ICLR, AAAI, IJCAI, UAI, and AISTATS
- 8/2019 **Invited Talk**, ITCS, Shanghai University of Finance & Economics
 Topic: *Preference Learning from General Partial Orders*
- 9/2019 – 5/2022 **RPI-IBM AI Horizon Scholarship** (fund 3 years' tuition and stipend)
- 9/2016 – 5/2017 **Presidential Graduate Research Fellowship** [[Certificate](#)]